

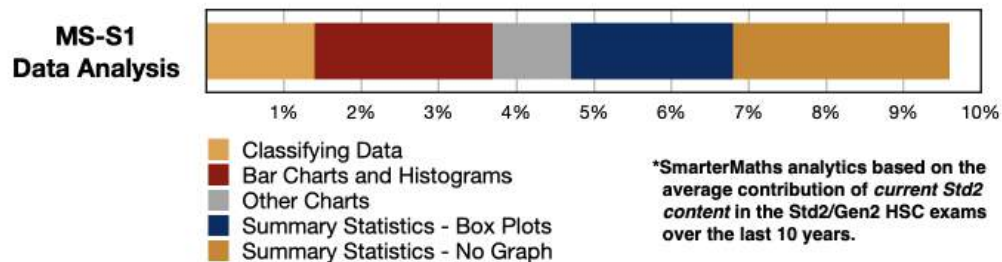
STD 1: Statistical Analysis (Std 1), S1 Data Analysis (Y11)

Summary Statistics - No Graph (Std 1)

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Exam Equivalent Time: 73.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

Std2/Gen2 Exam Contribution History S1 Data Analysis



IMPORTANT FEATURES AND TIPS FROM EXAM HISTORY

- MS-S1 Data Analysis has accounted for 6% and 5% in the 2020 and 2019 Std1 exams respectively.
- MS-S1 Data Analysis is split into the sub-topics listed in the graph above. Note that estimated exam contributions in the graph are taken from Std2/Gen2 past exams due to the very limited Std1 exam history. They should only be used for estimating the "relative" importance of each sub-topic.
- This specific analysis looks at the minor sub-topic *Summary Statistics - No Graph*.

ANALYSIS - What to Expect and Common pitfalls

- *Summary Statistics - No Graph* questions require students to understand and calculate statistics such as *median*, *mean* and *standard deviation* (by calculator) given a simple data set.
- This topic area attracted a significant 4-mark allocation in the 2020 Std1 exam (Q24), producing mean marks of 39% and 20% in its two parts. Revision attention required here!
- A core competency of calculating a five number summary from a data set is a must, along with understanding mean and median changes when a dataset is adjusted. Numerous database questions address these areas.
- The 2019 Std2 exam tested students on "outlier" calculations after the 2017 (Q30a) and 2015 (Q27d) exams exposed a lack of understanding in this area.
- *Marker Comments* have highlighted past issues for many students in finding the mean of grouped data, where they must use the "class centres" for their calculations. This caused major issues in both 2014 and 2006 and should be reviewed.

Questions

1. Statistics, STD2 S1 2005 HSC 1 MC

What is the mean of the set of scores?

3, 4, 5, 6, 6, 8, 8, 8, 15

- (A) 6
(B) 7
(C) 8
(D) 9

2. Statistics, STD2 S1 2009 HSC 3 MC

The eye colours of a sample of children were recorded.

When analysing this data, which of the following could be found?

- (A) Mean
(B) Median
(C) Mode
(D) Range

3. Statistics, STD2 S1 2018 HSC 1 MC

A set of scores has the following five-number summary.

lower extreme = 2
lower quartile = 5
median = 6
upper quartile = 8
upper extreme = 9

What is the range?

- A. 2
B. 3
C. 6
D. 7

4. Statistics, STD2 S1 2004 HSC 6-7 MC

Use the set of scores 1, 3, 3, 3, 4, 5, 7, 7, 12 to answer Questions 6 and 7.

Question 6

What is the range of the set of scores?

- (A) 6
- (B) 9
- (C) 11
- (D) 12

Question 7

What are the median and the mode of the set of scores?

- (A) Median 3, mode 5
 - (B) Median 3, mode 3
 - (C) Median 4, mode 5
 - (D) Median 4, mode 3
-

5. Statistics, STD2 S1 2011 HSC 14 MC

A data set of nine scores has a median of 7.

The scores 6, 6, 12 and 17 are added to this data set.

What is the median of the data set now?

- (A) 6
 - (B) 7
 - (C) 8
 - (D) 9
-

6. Statistics, STD2 S1 2011 HSC 17 MC

The heights of the players in a basketball team were recorded as 1.8 m, 1.83 m, 1.84 m, 1.86 m and 1.92 m. When a sixth player joined the team, the average height of the players increased by 1 centimetre.

What was the height of the sixth player?

- (A) 1.85 m
 - (B) 1.86 m
 - (C) 1.91 m
 - (D) 1.93 m
-

7. Statistics, STD2 S1 2013 HSC 14 MC

The July sales prices for properties in a suburb were:

\$552 000, \$595 000, \$607 000, \$607 000, \$682 000, and \$685 000.

On 1 August, another property in the same suburb was sold for over one million dollars.

If the property had been sold in July, what effect would it have had on the mean and median sale prices for July?

- (A) Both the mean and median would have changed.
 - (B) Neither the mean nor the median would have changed.
 - (C) The mean would have changed and the median would have stayed the same.
 - (D) The mean would have stayed the same and the median would have changed.
-

8. Statistics, STD2 S1 2008 HSC 13 MC

The height of each student in a class was measured and it was found that the mean height was 160 cm.

Two students were absent. When their heights were included in the data for the class, the mean height did not change.

Which of the following heights are possible for the two absent students?

- (A) 155 cm and 162 cm
 - (B) 152 cm and 167 cm
 - (C) 149 cm and 171 cm
 - (D) 143 cm and 178 cm
-

9. Statistics, STD2 S1 2008 HSC 8 MC

What is the median of the following set of scores?

<i>Score</i>	<i>Frequency</i>
12	13
14	6
16	2
18	12
Total	33

- (A) 12
 - (B) 13
 - (C) 14
 - (D) 15
-

10. Statistics, STD2 S1 2006 HSC 12 MC

The mean of a set of 5 scores is 62.

What is the new mean of the set of scores after a score of 14 is added?

- (A) 38
 - (B) 54
 - (C) 62
 - (D) 76
-

11. Statistics, STD2 S1 2018 HSC 11 MC

A set of data is summarised in this frequency distribution table.

<i>Score</i>	<i>Frequency</i>
3	1
4	2
5	6
6	7
7	9
8	5
Total = 30	

Which of the following is true about the data?

- A. Mode = 7, median = 5.5
 - B. Mode = 7, median = 6
 - C. Mode = 9, median = 5.5
 - D. Mode = 9, median = 6
-

12. Statistics, STD2 S1 2007 HSC 21 MC

This set of data is arranged in order from smallest to largest.

5, 6, 11, x , 13, 18, 25

The range is six less than twice the value of **x** .

Which one of the following is true?

- (A) The median is 12 and the interquartile range is 7.
 - (B) The median is 12 and the interquartile range is 12.
 - (C) The median is 13 and the interquartile range is 7.
 - (D) The median is 13 and the interquartile range is 12.
-

13. Statistics, STD2 S1 2014 HSC 14 MC

Twenty Year 12 students were surveyed. These students were asked how many hours of sport they play per week, to the nearest hour.

The results are shown in the frequency table.

<i>Hours per week</i>	<i>Frequency</i>
0–2	5
3–5	10
6–8	3
9–11	2

What is the mean number of hours of sport played by the students per week?

- (A) 3.3
 - (B) 4.3
 - (C) 5.0
 - (D) 5.3
-

14. Statistics, STD2 S1 2016 HSC 19 MC

A soccer referee wrote down the number of goals scored in 9 different games during the season.

2, 3, 3, 3, 5, 5, 8, 9, ...

The last number has been omitted. The range of the data is 10.

What is the five-number summary for this data set?

- (A) 2, 3, 5, 8.5, 12
 - (B) 2, 3, 5, 8.5, 10
 - (C) 2, 3, 5, 8, 12
 - (D) 2, 3, 5, 8, 10
-

15. Statistics, STD2 S1 2016 HSC 21 MC

A grouped data frequency table is shown.

<i>Class interval</i>	<i>Frequency</i>
1–5	3
6–10	6
11–15	8
16–20	9

What is the mean for this set of data?

- (A) 6.5
 - (B) 10.5
 - (C) 11.9
 - (D) 12.4
-

16. Statistics, STD2 S1 2009 HSC 21 MC

The mean of a set of ten scores is 14. Another two scores are included and the new mean is 16.

What is the mean of the two additional scores?

- (A) 4
 - (B) 16
 - (C) 18
 - (D) 26
-

17. Statistics, STD2 S1 2007 HSC 24a

Consider the following set of scores:

3, 5, 5, 6, 8, 8, 9, 10, 10, 50.

- i. Calculate the mean of the set of scores. *(1 mark)*
 - ii. What is the effect on the mean and on the median of removing the outlier? *(2 marks)*
-

18. Statistics, STD2 S1 2013 HSC 26b

Write down a set of six data values that has a range of 12, a mode of 12 and a minimum value of 12.
(2 marks)

19. Statistics, STD2 S1 2021 HSC 17

The five-number summary of a dataset is given.

Lowest score = 1

Lowest quartile (Q_1) = 4

Median (Q_2) = 7

Upper quartile (Q_3) = 10

Highest score = 20

Is 20 an outlier? Justify your answer with calculations. (2 marks)

20. Statistics, STD2 S1 2017 HSC 27a

Jamal surveyed eight households in his street. He asked them how many kilolitres (kL) of water they used in the last year. Here are the results.

220, 105, 101, 450, 37, 338, 151, 205

i. Calculate the mean of this set of data. (1 mark)

ii. What is the standard deviation of this set of data, correct to one decimal place? (1 mark)

21. Statistics, STD2 S1 2006 HSC 23c

Vicki wants to investigate the number of hours spent on homework by students at her high school.

i. Briefly describe a valid method of randomly selecting 200 students for a sample. (1 mark)

ii. Vicki chooses her sample and asks each student how many hours (to the nearest hour) they usually spend on homework during one week.

The responses are shown in the frequency table.

Number of hours spent on homework in a week	Frequency
0 to 4	69
5 to 9	72
10 to 14	38
15 to 19	21

What is the mean amount of time spent on homework? (2 marks)

22. Statistics, STD2 S1 2015 HSC 27d

In a small business, the seven employees earn the following wages per week:

\$300, \$490, \$520, \$590, \$660, \$680, \$970.

i. Is the wage of \$970 an outlier for this set of data? Justify your answer with calculations. (3 marks)

ii. Each employee receives a \$20 pay increase.

What effect will this have on the standard deviation? (1 mark)

23. Statistics, STD2 S1 2008 HSC 23f

Christina has completed three Mathematics tests. Her mean mark is 72%.

What mark (out of 100) does she have to get in her next test to increase her mean mark to 73%? (2 marks)

24. Statistics, STD2 S1 2017 HSC 30a

A set of data has a lower quartile (Q_L) of 10 and an upper quartile (Q_U) of 16.

What is the maximum possible range for this set of data if there are no outliers? (2 marks)

25. Statistics, STD2 S1 2018 HSC 26e

A cumulative frequency table for a data set is shown.

Score	Cumulative frequency
1	5
2	9
3	16
4	20
5	34
6	42

What is the interquartile range of this data set? (2 marks)

26. Statistics, STD1 S1 2019 HSC 19

The heights, in centimetres, of 10 players on a basketball team are shown.

170, 180, 185, 188, 192, 193, 194, 196, 202

Is the height of the shortest player on the team considered an outlier? Justify your answer with calculations. (3 marks)

27. Statistics, STD1 S1 2020 HSC 24

a. The ages in years, of ten people at the local cinema last Saturday afternoon are shown.

38 25 38 46 55 68 72 55 36 38

The mean of this dataset is 47.1 years.

How many of the ten people were aged between the mean age and the median age? (2 marks)

b. On Wednesday, ten people all aged 70 went to this same cinema.

Would the standard deviation of the age dataset from Wednesday be larger than, smaller than or equal to the standard deviation of the age dataset given in part (a)? Briefly explain your answer without performing any calculations. (2 marks)

28. Statistics, STD2 S1 2005 HSC 27d

Nine students were selected at random from a school, and their ages were recorded.

Ages		
12	11	16
14	16	15
14	15	14

i. What is the sample standard deviation, correct to two decimal places? (2 marks)

ii. Briefly explain what is meant by the term *standard deviation*. (1 mark)

Worked Solutions

1. Statistics, STD2 S1 2005 HSC 1 MC

$$\begin{aligned}\text{Mean} &= \frac{(3 + 4 + 5 + 6 + 6 + 8 + 8 + 8 + 15)}{9} \\ &= \frac{63}{9} \\ &= 7 \\ &\Rightarrow B\end{aligned}$$

2. Statistics, STD2 S1 2009 HSC 3 MC

Eye colour is categorical data

$\therefore$ Only the mode can be found

$\Rightarrow C$

3. Statistics, STD2 S1 2018 HSC 1 MC

$$\begin{aligned}\text{Range} &= \text{upper extreme} - \text{lower extreme} \\ &= 9 - 2 \\ &= 7 \\ &\Rightarrow D\end{aligned}$$

4. Statistics, STD2 S1 2004 HSC 6-7 MC

Question 6

$$\text{Range} = \text{High} - \text{Low}$$

$$= 12 - 1$$

$$= 11$$

$$\Rightarrow C$$

Question 7

9 scores

$$\therefore \text{Median} = \frac{9 + 1}{2}$$

$$= 5\text{th score}$$

$$= 4$$

$$\text{Mode} = 3$$

$$\Rightarrow D$$

5. Statistics, STD2 S1 2011 HSC 14 MC

Since an even amount of scores are added below and above the existing median, it will not change.

$$\Rightarrow B$$

6. Statistics, STD2 S1 2011 HSC 17 MC

$$\text{Old Mean} = (1.8 + 1.83 + 1.84 + 1.86 + 1.92) \div 5$$

$$= \frac{9.25}{5}$$

$$= 1.85 \text{ m}$$

Since the new mean = 1.86m (given)

$$\text{New Mean} = \text{Height of all 6 players} \div 6$$

$$\therefore 1.86 = \frac{9.25 + h}{6} \quad (h = \text{height of new player})$$

$$h = (6 \times 1.86) - 9.25$$

$$= 1.91 \text{ m}$$

$$\Rightarrow C$$

7. Statistics, STD2 S1 2013 HSC 14 MC

Mean increases because new house is sold above the existing average.

$$\text{Initial median} = \frac{607\,000 + 607\,000}{2} = 607\,000$$

$$\text{New median} = 607\,000 \quad (4\text{th value in a list of 7})$$

$$\Rightarrow C$$

8. Statistics, STD2 S1 2008 HSC 13 MC

Since the mean doesn't change

$$\Rightarrow 2 \text{ absent students must have a mean height of } 160 \text{ cm.}$$

Considering each option given,

$$(149 + 171) \div 2 = 160$$

$$\Rightarrow C$$

9. Statistics, STD2 S1 2008 HSC 8 MC

$$\text{Median} = \frac{n + 1}{2}$$

$$= \frac{33 + 1}{2}$$

$$= 17\text{th score}$$

$$\therefore \text{Median is } 14$$

$$\Rightarrow C$$

10. Statistics, STD2 S1 2006 HSC 12 MC

$$\text{Mean of 5 scores} = 62$$

$$\therefore \text{Total of 5 scores} = 62 \times 5 = 310$$

Add a score of 14

$$\text{Total of 6 scores} = 310 + 14 = 324$$

$$\therefore \text{New mean} = \frac{324}{6}$$

$$= 54$$

$$\Rightarrow B$$

11. Statistics, STD2 S1 2018 HSC 11 MC

Mode = 7 (highest frequency of 9)

Median = average of 15th and 16th data points.

$$\therefore \text{Median} = 6$$

$\Rightarrow B$

12. Statistics, STD2 S1 2007 HSC 21 MC

5, 6, 11, x , 13, 18, 25

$$\text{Range} = 2x - 6$$

$$25 - 5 = 2x - 6$$

$$2x = 26$$

$$x = 13$$

$$\therefore \text{Median} = 13$$

$$Q_1 = 6 \quad Q_3 = 18$$

$$\therefore \text{IQR} = 12$$

$\Rightarrow D$

13. Statistics, STD2 S1 2014 HSC 14 MC

Using the class centres

$$\text{Total hours} = (1 \times 5) + (4 \times 10) + (7 \times 3) + (10 \times 2)$$

$$= 5 + 40 + 21 + 20$$

$$= 86$$

$$\text{Mean hours} = \frac{86}{20} = 4.3$$

$\Rightarrow B$

♦ Mean mark 45%
COMMENT: The mean is calculated using "class centres" in grouped data.

14. Statistics, STD2 S1 2016 HSC 19 MC

Since range is 10,

Last data point = 12

$$Q_1 = 3$$

$$Q_3 = \frac{8 + 9}{2} = 8.5$$

Median = 5

$\Rightarrow A$

♦ Mean mark 46%.

15. Statistics, STD2 S1 2016 HSC 21 MC

Using the centre of each class interval:

$$\text{Mean} = \frac{3 \times 3 + 8 \times 6 + 13 \times 8 + 18 \times 9}{3 + 6 + 8 + 9}$$

$$= 12.42 \dots$$

$\Rightarrow D$

♦ Mean mark 43%.

16. Statistics, STD2 S1 2009 HSC 21 MC

If $\bar{x}$ of 10 scores = 14

$$\Rightarrow \text{Sum of 10 scores} = 10 \times 14 = 140$$

With 2 additional scores, $\bar{x} = 16$

$$\Rightarrow \text{Sum of 12 scores} = 12 \times 16 = 192$$

$$\therefore \text{Value of 2 extra scores} = 192 - 140 = 52$$

$$\therefore \text{Mean of 2 extra scores} = \frac{52}{2} = 26$$

$\Rightarrow D$

♦♦ Mean mark 28%.

17. Statistics, STD2 S1 2007 HSC 24a

i. Total of scores

$$= 3 + 5 + 5 + 6 + 8 + 8 + 9 + 10 + 10 + 50$$

$$= 114$$

$$\therefore \text{Mean} = \frac{114}{10} = 11.4$$

ii. Mean

If the outlier (50) is removed, the mean would become lower.

Median

The current median (10 data points)

$$= \frac{5\text{th} + 6\text{th}}{2} = \frac{8 + 8}{2} = 8$$

The new median (9 data points)

$$= 5\text{th value}$$

$$= 8$$

$\therefore$ Median will NOT change.

18. Statistics, STD2 S1 2013 HSC 26b

12, 12, 12, 16, 18, 24

NB. There are many correct solutions.

19. Statistics, STD2 S1 2021 HSC 17

$$IQR = Q_3 - Q_1 = 10 - 4 = 6$$

$$\text{Upper Fence} = Q_3 + 1.5 \times IQR$$

$$= 10 + 1.5 \times 6$$

$$= 19$$

Since $20 > 19$, 20 is an outlier.

COMMENT: Stating $20 > 19$ is necessary to justify this answer.

20. Statistics, STD2 S1 2017 HSC 27a

i. Mean = $(220 + 105 + 101 + 450 + 37 + 338 + 151 + 205) \div 8$

$$= 200.875$$

ii. Std Dev = 127.357... (by calc)

$$= 127.4 \text{ (1 d.p.)}$$

♦ Mean mark part (ii) 47%.
IMPORTANT: The population standard deviation is required here.

21. Statistics, STD2 S1 2006 HSC 23c

i. A valid method would be using a stratified sample.

The number of students sampled in each year is proportional to the size of each year.

ii.

Hours	Class Centre (x)	Frequency (f)	fx
0 - 4	2	69	138
5 - 9	7	72	504
10 - 14	12	38	456
15 - 19	17	21	357
Total			1455

MARKER'S COMMENT: This "routine" exercise of finding a mean from grouped data was incorrectly answered by most students! The best responses copied the table and inserted a class-centre column (see solution).

$$\text{Mean} = \frac{\text{Sum of Scores}}{\text{Total scores}}$$

$$= \frac{1455}{200}$$

$$= 7.275 \text{ hours}$$

22. Statistics, STD2 S1 2015 HSC 27d

i. 300, 490, 520, 590, 660, 680, 970

$$\text{Median} = 590$$

$$Q_1 = 490$$

$$Q_3 = 680$$

$$\therefore IQR = 680 - 490 = 190$$

Outlier if value is greater than

$$Q_3 + 1.5 \times IQR$$

$$= 680 + 1.5 \times 190$$

$$= 965$$

$\therefore$ The wage \$970 is an outlier.

♦ Mean mark 39%.

ii. All values increase by \$20, but so too does

the mean. Therefore the spread about the new mean will not change and therefore the standard deviation will remain the same.

23. Statistics, STD2 S1 2008 HSC 23f

Total marks in 3 tests

$$= 3 \times 72$$

$$= 216$$

We need 4-test mean = 73

i.e. Total Marks (4 tests) $\div 4 = 73$

$$\text{Total Marks (4 tests)} = 292$$

$$\therefore \text{4th test score} = 292 - 216 \\ = 76$$

24. Statistics, STD2 S1 2017 HSC 30a

$$IQR = 16 - 10 = 6$$

If no outliers,

$$\text{Upper limit} = Q_U + 1.5 \times IQR$$

$$= 16 + 1.5 \times 6$$

$$= 25$$

$$\text{Lower limit} = Q_L - 1.5 \times IQR$$

$$= 10 - 1.5 \times 6$$

$$= 1$$

$$\therefore \text{Maximum range} = 25 - 1 \\ = 24$$

♦♦ Mean mark 34%.

25. Statistics, STD2 S1 2018 HSC 26e

$$42 \text{ data points} \Rightarrow \text{median} = \frac{21\text{st} + 22\text{nd}}{2}$$

$$Q_1 = 11\text{th data point} = 3$$

$$Q_3 = 32\text{nd data point} = 5$$

$$\therefore IQR = 5 - 3 \\ = 2$$

♦♦ Mean mark 27%.

26. Statistics, STD1 S1 2019 HSC 19

$$Q_1 = 185, Q_3 = 194$$

$$IQR = 194 - 185 = 9$$

Shortest player = 170

Outlier height:

$$Q_1 - 1.5 \times IQR = 185 - 1.5 \times 9 \\ = 171.5$$

$\therefore$ Since $170 < 171.5$, 170 is an outlier.

♦♦ Mean mark 24%.

COMMENT: The last statement must be made to achieve full marks here!

27. Statistics, STD1 S1 2020 HSC 24

a. In ascending order:

25, 36, 38, 38, 38, 46, 55, 55, 68, 72

◆ Mean mark part (a) 39%.

$$\text{Median} = \frac{5\text{th} + 6\text{th}}{2} = \frac{38 + 46}{2} = 42$$

∴ People with $42 < \text{age} < 47.1 = 1$

b. Standard deviation is a measure of how much the

◆◆ Mean mark part (b) 20%.

ages of individuals differ from the mean age of the group.

∴ Standard deviation of Wednesday group would be less as the mean is 70 and everyone's age is 70.

28. Statistics, STD2 S1 2005 HSC 27d

i. Sample standard deviation

= 1.6914... (by calculator)

= 1.69 (to 2 d.p.)

ii. Standard deviation is a measure of how much members of a data group differ from the mean value of the group.